

SECURE BOOT, FIRMWARE INTEGRITY, AUTHENTICATION, AUDITABILITY, AND DATA LIFECYCLE CONTROLS (NON-LIMITING)

A. Secure Boot and Firmware Integrity (Non-Limiting)

In various embodiments, each node implements protections to reduce tampering and unauthorized firmware modification. Non-limiting approaches include: 1. secure boot, wherein execution is restricted to firmware images that are cryptographically verified; 2. signed firmware updates, wherein update packages are verified prior to installation; 3. firmware version checks and rollback prevention; and 4. integrity checks for configuration parameters and calibration tables. In certain embodiments, firmware authenticity is verified using asymmetric signatures and device-stored trust anchors.

B. Device Authentication and Mutual Trust (Non-Limiting)

In various embodiments, a node authenticates to a user device and/or computing system using device identity credentials. Non-limiting mechanisms include: 1. device certificates or device keys provisioned at manufacturing or first initialization; 2. mutual authentication during session establishment; 3. ephemeral session keys derived per connection; and 4. revocation and re-pairing flows when compromise is suspected. In certain embodiments, the node refuses data transfer if authentication is not satisfied.

C. Application-Layer Authorization and Role Boundaries (Non-Limiting)

In various embodiments, authorization controls separate: 1. on-device control actions (e.g., stimulus control for session-based devices); 2. access to raw sensor streams; and 3. access to derived privacy-preserving metrics. In certain embodiments, the system enforces least-privilege boundaries such that a user-facing application can access indices and confidence indicators without accessing raw high-resolution signals unless explicitly enabled.

D. Auditability and Security Event Logging (Non-Limiting)

In various embodiments, the node and/or associated computing system generates security-relevant logs including, without limitation: 1. pairing/bonding events; 2. firmware update events; 3. failed authentication attempts; 4. changes to consent gating states; and 5. secure erase events. In certain embodiments, logs are stored in encrypted form and are exportable only under explicit user action.

E. Data Lifecycle Controls, Minimization, and Secure Erase (Non-Limiting)

In various embodiments, the system enforces data minimization and lifecycle policies including: 1. buffering raw data only for the duration needed to compute derived features; 2. automatic deletion of raw buffers after successful synchronization or after a defined retention window; 3. secure erase of cached records (e.g., cryptographic erase via key destruction); and 4. user-initiated wipe flows for loss scenarios. In certain embodiments, derived metrics are stored at lower sensitivity than raw sensor streams and may be retained longer under user settings.

F. Cloud and Server-Side Security (Optional, Non-Limiting)

In certain embodiments, when remote computing is used, server-side controls include: 1. encryption at rest and in transit; 2. access controls and separation of duties; 3. key management systems; and 4. monitoring for anomalous access patterns. In certain embodiments, raw data is not uploaded unless explicitly enabled,

and default behaviors favor local computation and privacy-preserving summaries.

G. Cross-Node Correlation Under Consent (Non-Limiting)

In various embodiments, correlation outputs across nodes require explicit consent gating and may be computed using: 1. time-aligned windows passing quality thresholds; 2. privacy-preserving feature representations; and 3. confidence-weighted scores. In certain embodiments, cross-node outputs are generated without exposing raw underlying signals across nodes.

H. Variations

Any embodiment described herein may use different cryptographic primitives, authentication models, logging schemes, and retention policies, while maintaining the functional intent of tamper resistance, authenticated access, secure update pathways, and minimized data exposure.